



6.3.5 Matrix-matrix multiplication ¶

6.3.5 Error in matrix matrix multiplication, Part 1

fit width



The idea behind backward error analysis is that the computed result is the exact result when computing with changed inputs. Let's consider matrix-matrix multiplication:

$$C := AB.$$

What we would like to be able to show is that there exist ΔA and ΔB such that the computed result, $\tilde{C}$, satisfies

$$\tilde{C} := (A + \Delta A)(B + \Delta B).$$

Let's think about this...

Ponder This 6.3.5.1. Can one find matrices ΔA and ΔB such that

$$\tilde{C} = (A + \Delta A)(B + \Delta B)?$$

6.3.5 Error in matrix matrix multiplication, Part 2

fit width



For matrix-matrix multiplication, it is possible to "throw" the error onto the result, as summarized by the following theorem:

Theorem 6.3.5.1. Forward error for matrix-matrix multiplication.
Let $C \in \mathbb{R}^{m \times n}$, $A \in \mathbb{R}^{m \times k}$, and $B \in \mathbb{R}^{k \times n}$ and consider the assignment $C := AB$ implemented via matrix-vector multiplication. Then there exists $\Delta C \in \mathbb{R}^{m \times n}$ such that

$$\tilde{C} = AB + \Delta C, \text{ where } |\Delta C| \leq \gamma_k |A| |B|.$$

Homework 6.3.5.2. Prove [Theorem 6.3.5.1](#).

▼ [Solution](#)

Partition

$$C = \left(\begin{array}{c|c|c|c} c_0 & c_1 & \cdots & c_{n-1} \end{array} \right) \quad \text{and} \quad B = \left(\begin{array}{c|c|c|c} b_0 & b_1 & \cdots & b_{n-1} \end{array} \right).$$

Then

$$\left(\begin{array}{c|c|c|c} \tilde{c}_0 & \tilde{c}_1 & \cdots & \tilde{c}_{n-1} \end{array} \right) := \left(\begin{array}{c|c|c|c} Ab_0 & Ab_1 & \cdots & Ab_{n-1} \end{array} \right).$$

From [R-1F 6.3.4.1](#) regarding matrix-vector multiplication we know that

$$\begin{aligned} \left(\begin{array}{c|c|c|c} \tilde{c}_0 & \tilde{c}_1 & \cdots & \tilde{c}_{n-1} \end{array} \right) &= \left(\begin{array}{c|c|c|c} Ab_0 + \delta_0 & Ab_1 + \delta_1 & \cdots & Ab_{n-1} + \delta_{n-1} \end{array} \right) \\ &= \left(\begin{array}{c|c|c|c} Ab_0 & Ab_1 & \cdots & Ab_{n-1} \end{array} \right) + \left(\begin{array}{c|c|c|c} \delta_0 & \delta_1 & \cdots & \delta_{n-1} \end{array} \right) \\ &= AB + \Delta C. \end{aligned}$$

where $|\delta_j| \leq \gamma_k |A| |b_j|$, $j = 0, \dots, n-1$, and hence $|\Delta C| \leq \gamma_k |A| |B|$.

6.3.5 Error in matrix matrix multiplication, Part 3

fit width



Remark 6.3.5.2. In practice, matrix-matrix multiplication is often the parameterized operation $C := \alpha AB + \beta C$. A consequence of [Theorem 6.3.5.1](#) is that for $\beta \neq 0$, the error *can* be attributed to a change in parameter C , which means the error has been "thrown back" onto an input parameter.